

# Project Plan — Smart Scientific Calculator

## 1. Project Overview

| Field             | Detail                                                                                          |
|-------------------|-------------------------------------------------------------------------------------------------|
| Project Title     | Smart Scientific Calculator                                                                     |
| Development Model | Incremental Software Development Model                                                          |
| Technology Stack  | HTML, CSS, JavaScript (Frontend) · Node.js, Express (Backend) · MongoDB via Mongoose (Database) |
| Team Size         | [Fill in: e.g. 4 members]                                                                       |
| Project Duration  | 1 week (7 days)                                                                                 |
| Start Date        | [Fill in]                                                                                       |
| Mid-Sem Deadline  | 3 October 2026                                                                                  |
| Final Deadline    | End of Day 7 (Week 1)                                                                           |

## 2. Development Iterations & Timeline

The entire project is planned for a 1-week duration (Day 1 to Day 7). It follows six incremental iterations, each adding a well-defined set of features on top of the previous increment. Implementation (Iterations 1–6) is completed by Day 5, followed by testing, documentation and polish on Days 6–7.

| Iteration | Feature Area                                                                         | Planned Start | Planned End | Status      |
|-----------|--------------------------------------------------------------------------------------|---------------|-------------|-------------|
| 1         | Basic Arithmetic (UN-01): +, -, *, /, %, AC, DEL                                     | Day 1         | Day 1       | Completed   |
| 2 & 3     | Scientific Functions (UN-02 & UN-03): $\sqrt{\phantom{x}}$ , sin, cos, $x^y$         | Day 2         | Day 2       | Completed   |
| 4         | Error Handling (UN-04): Division by zero, invalid domain, syntax errors              | Day 3         | Day 3       | Completed   |
| 5         | UI & Interaction (UN-05): Responsive grid layout, real-time display, button feedback | Day 4         | Day 4       | Completed   |
| 6         | History & Database (UN-06): Express server, MongoDB CRUD, history view               | Day 4         | Day 5       | Completed   |
| —         | Testing, Documentation & Final Polish                                                | Day 6         | Day 7       | In Progress |

## 3. Milestone Tracker

| Milestone                         | Target Date  | Status |
|-----------------------------------|--------------|--------|
| Problem statement finalised       | Day 1        | Done   |
| Requirements (SRS) document ready | Day 1        | Done   |
| Architecture & DB design approved | Day 1        | Done   |
| Iteration 1 demo-ready            | End of Day 1 | Done   |
| Iteration 2 & 3 demo-ready        | End of Day 2 | Done   |
| Iteration 4 demo-ready            | End of Day 3 | Done   |
| Iteration 5 demo-ready            | End of Day 4 | Done   |
| Iteration 6 demo-ready            | End of Day 5 | Done   |

| Milestone               | Target Date | Status      |
|-------------------------|-------------|-------------|
| Mid-semester submission | 3 Oct 2026  | In Progress |
| Final testing & polish  | Day 6       | Planned     |
| Final submission        | Day 7       | Planned     |

## 4. Task Allocation Table

| Task                            | Assigned To            | Status      |
|---------------------------------|------------------------|-------------|
| Requirements gathering & SRS    | [Member 1], [Member 2] | Done        |
| Use-case diagram / User stories | [Member 2]             | Done        |
| Architecture & database design  | [Member 3]             | Done        |
| UI wireframe & prototype        | [Member 4]             | Done        |
| Iteration 1 implementation      | [Member 1], [Member 3] | Done        |
| Iteration 2 & 3 implementation  | [Member 2], [Member 4] | Done        |
| Iteration 4 implementation      | [Member 1]             | Done        |
| Iteration 5 implementation      | [Member 4]             | Done        |
| Iteration 6 implementation      | [Member 3]             | Done        |
| Test-case design & execution    | [Member 2], [Member 4] | Done        |
| Mid-semester report writing     | All members            | In Progress |
| Demonstration video             | [Member 1], [Member 3] | Planned     |

Note: Replace [Member N] with actual team-member names.

## 5. Risks and Mitigation

| # | Risk                                             | Likelihood | Impact | Mitigation Strategy                                           |
|---|--------------------------------------------------|------------|--------|---------------------------------------------------------------|
| 1 | MongoDB unavailable on deployment machine        | Medium     | High   | Provide Docker Compose file; document manual setup steps      |
| 2 | eval() in front-end poses security risk          | Medium     | Medium | Planned: replace with a safe expression parser in final phase |
| 3 | Floating-point precision in scientific functions | Low        | Low    | Document known tolerance; round display to 10 decimal places  |
| 4 | Team-member unavailability near deadline         | Low        | High   | Maintain shared repo; any member can pick up pending work     |
| 5 | Moodle upload size limit exceeded                | Low        | Medium | Host demo video externally; include DEMO_VIDEO_LINK.txt       |

## 6. Delayed or Pending Tasks

| Task                                | Original Target | Revised Target | Reason for Delay           |
|-------------------------------------|-----------------|----------------|----------------------------|
| Expression parser to replace eval() | Day 5           | Post mid-sem   | Prioritised DB integration |
| Keyboard input support              | Day 4           | Post mid-sem   | Scope deferred             |
| Unit tests with Jest (automated)    | Day 6           | Post mid-sem   | Manual testing done first  |

## 7. Work Planned for Remaining Semester

1. Replace eval() with a secure mathematical expression parser.

2. Add keyboard input support for desktop users.
3. Implement automated unit tests using Jest.
4. Add a proper in-page history panel (instead of browser alert()).
5. Deploy the application to a cloud platform (e.g. Render / Railway).
6. Conduct usability testing with peer users.
7. Final documentation and submission packaging.